

Cryptography Basics and Applications

Symmetric Encryption

Uses same key for encryption and decryption. Common algorithms: AES (most widely used), DES (insecure), 3DES. AES features 128-bit block size with key lengths of 128/192/256 bits.

Asymmetric Encryption

Uses public key and private key. RSA: based on integer factorization, typically 2048+ bits. ECC: elliptic curve cryptography with shorter keys.

Hash Functions

Maps arbitrary length data to fixed-length hash. Common: SHA-256, SHA-384, SHA-512. MD5 and SHA-1 are insecure.